$$
\begin{aligned}
\pi &= (0.6, 0.3, 0.1), \quad p = (0.2, 0.5, 0.3) \\
\ell^p &= -[0.6 \log 0.2 + 0.3 \log 0.5 + 0.1 \log 0.3] \\
&\approx 1.294 \\
z_t &= 1 + 0.9 \times 0 + 0.9^2 \times 2 + 0.9^3 \times 5 \\
&= 6.265
\end{aligned}
$$